## Processing code

```
/*Ultrasonic radar system.
  *Run the following code a second time.
  *created by the SriTu Tech team.
  *Read the code below and use it for any of your creations.
*/

//import following libraries
Import processing.serial.*;
Import java.awt.event.KeyEvent;
Import java.io.IOException;

Serial myport;//create serial object

PFont f;
Int Angle, Distance;
String angle=””;
String distance=””;
String data;
Int index1=0;
Int index2=0;
Float pixsDistance;

Void setup() {
  Size(1250, 700);//screen size
  Smooth();
```

```
  printArray(PFont.list());//show font list your computer
  f = createFont("David Bold", 30);//font name and size
  textFont(f);

  String portName = Serial.list()[0];//set COM port
  Myport = new Serial(this, portName, 9600);
  Myport.bufferUntil('.');
}

Void draw() {

  noStroke();
  fill(0, 10);
  rect(0, 0, width, 700);
  fill(98, 245, 31);
  greenmesh();
  radararea();
  words();
  greenLine();
  redline();
}
//get ardunio board serial values
Void serialEvent (Serial myport) {
  Data = myport.readStringUntil('.');
  Data = data.substring(0, data.length()-1);
```

```
  Index1 = data.indexOf(“,”);
  Angle= data.substring(0, index1);
  Distance= data.substring(index1+1, data.length());

  // converts the String variables into Integer
  Angle = int(angle);
  Distance = int(distance);
}

//Half circle and lines
Void radararea() {
  pushMatrix();
  translate(625, 680);
  noFill();
  strokeWeight(2);
  stroke(98, 245, 31);
  // draws the arc lines
  Arc(0, 0, 1150, 1150, PI, TWO_PI);
  Arc(0, 0, 850, 850, PI, TWO_PI);
  Arc(0, 0, 550, 550, PI, TWO_PI);
  Arc(0, 0, 250, 250, PI, TWO_PI);
  // draws the angle lines
  Line(-450, 0, 450, 0);
  Line(0, 0, -600*cos(radians(30)), -600*sin(radians(30)));
  Line(0, 0, -600*cos(radians(60)), -600*sin(radians(60)));
  Line(0, 0, -600*cos(radians(90)), -600*sin(radians(90)));
```

```
  Line(0, 0, -600*cos(radians(120)), -600*sin(radians(120)));
  Line(0, 0, -600*cos(radians(150)), -600*sin(radians(150)));
  Line(-960*cos(radians(30)), 0, 960, 0);
  popMatrix();
}

//Green box net
Void greenmesh() {
  Stroke(98, 245, 31);
  strokeWeight(0.1);
  for (int x=0; x<=700; x+=5) {
    line(0, x, width, x);
  }
  For (int y=0; y<=1250; y+=5) {
    Line(y, 0, y, height);
  }
}

//print text
Void words() {
  Fill(98, 245, 31);
  Text(“180’”, 10, 670);
  Fill(98, 245, 31);
  Text(“0’”, 1210, 670);
  Fill(98, 245, 31);
  Text(“30’”, 1160, 380);
```

```
  Fill(98, 245, 31);
  Text("60'", 940, 160);
  Fill(98, 245, 31);
  Text("90'", 615, 70);
  Fill(98, 245, 31);
  Text("120'", 310, 150);
  Fill(98, 245, 31);
  Text("150'", 80, 370);
  Fill(255);
  Text("SriTu Tech Radar system", 20, 30);
  Fill(255);
  Text("Angle – "+Angle+" '", 20, 60);
  Fill(255);
  Text("Distance – "+Distance+" cm", 20, 90);
}
//Drawing green lines
Void greenLine() {
  pushMatrix();
  strokeWeight(7);
  stroke(30, 250, 60);//green color
  translate(625, 680);
  line(0, 0, 600*cos(radians(Angle)), -600*sin(radians(Angle)));
  popMatrix();
}
//Drawing red lines
Void redline() {
```

```
 pushMatrix();
 translate(625, 680);
 strokeWeight(7);
 stroke(255, 10, 10); //red color
 pixsDistance = Distance*22.5;
 // limiting the range to 40 cm
 If (Distance<40) {
   Line(pixsDistance*cos(radians(Angle)), -pixsDistance*sin(radians(Angle)),
600*cos(radians(Angle)), -600*sin(radians(Angle)));
 }
 popMatrix();
}
```